Written Assignment #1 Solutions

1 Work and Span

We use the binary-fork model. In particular, creating m parallel tasks by binary forking contributes $\Theta(\log m)$ span. Each call to `do_something()` still has constant work and constant span.

(1) There are $(n+1)^2 = \Theta(n^2)$ total calls to `do_something()`, so the work is

$$W(n) = \Theta(n^2).$$

For the span, the outer `parallel_for` creates $\Theta(n)$ iterations by binary forking, which costs $\Theta(\log n)$ span. Along any one outer iteration, the inner `parallel_for` also creates $\Theta(n)$ iterations and contributes another $\Theta(\log n)$ span. Hence

$$S(n) = \Theta(\log n) + \Theta(\log n) = \Theta(\log n).$$

(2) The work recurrence is

$$W(n) = 2W(n/8) + \Theta(1).$$

Here $n^{\log_8 2} = n^{1/3}$, so by the Master Theorem,

$$W(n) = \Theta(n^{1/3}).$$

The two recursive calls run in parallel, so the span satisfies

$$S(n) = S(n/8) + \Theta(1) = \Theta(\log n).$$

(3) The work recurrence is

$$W(n) = 2W(n/2) + \Theta(\sqrt{n}).$$

Since $n^{\log_2 2} = n$ and $\sqrt{n} = O(n^{1-\varepsilon})$, the recursive part dominates, giving

$$W(n) = \Theta(n).$$

The recursive calls run in parallel, so they contribute $S(n/2)$ span. The `parallel_for` over $\sqrt{n}$ iterations costs $\Theta(\log \sqrt{n}) = \Theta(\log n)$ span under binary fork. Therefore

$$S(n) = S(n/2) + \Theta(\log n).$$

Summing over the $\Theta(\log n)$ recursion levels gives

$$S(n) = \Theta((\log n)^2).$$

2 (Another) Parallel Scan Algorithm

1. Divide-and-conquer: $W(n) = O(n)$, $D(n) = O(\log n)$.
   Decrease-and-conquer: $W(n) = O(n)$, $D(n) = O(\log^2 n)$.

2. We can use sequential scan on $S[1..k]$ to compute $B[in/k]$.

3. For each chunk $A[(i-1)n/k + 1...in/k]$, we use sequential scan to compute the prefix sums $B[(i-1)n/k + 1...in/k]$. Then we sequentially add $B[(i-1)n/k]$ to each of $B[(i-1)n/k + 1...in/k]$.

4. Step 2: $W_2(n) = k \times n/k = O(n)$.
   Step 3: $W_3(n) = O(k)$.
   Step 4: $W_4(n) = k \times n/k = O(n)$.
   So $W(n) = W_1(n) + W_2(n) + W_3(n) = O(n)$.

5. Step 2: $D_2(n) = O(\log k) + O(\log(n/k)) = O(\log n)$.
   Step 3: $D_3(n) = O(k)$.
   Step 4: $D_4(n) = O(\log k) + O(n/k)$.
   So $D(n) = D_1(n) + D_2(n) + D_3(n) = O(\log n) + O(k) + O(n/k)$.

6. When $k = \sqrt{n}$, we have the minimum span $O(\sqrt{n})$.